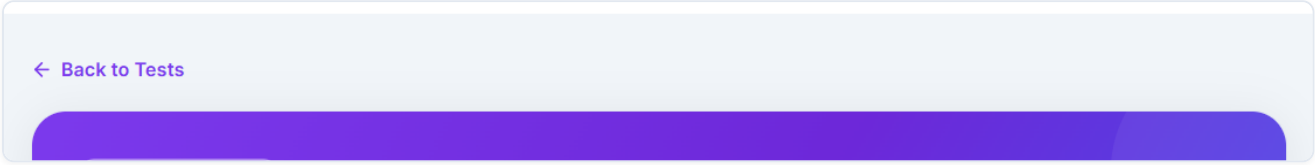
🏆 Infosys SE Full-Length Mock Exam 06 — Actual Exam Simulation

Official Proctored Examination | Date: October 29, 2026 | LearnSquare LMS

Conducted on: October 29, 2026 | **Session**: 4:30 PM – 6:00 PM | **Total Questions**: 62 Questions | **Total Duration**: 90 Minutes | **Total Marks**: 90 Marks

📌 Executive Exam Architecture & Scoring Rubric



This master compendium documents the complete, unedited, authentic **Infosys Pattern Test - 1** conducted on **LearnSquare LMS** on **October 29, 2026**. Every single problem statement, multiple-choice option set, public and private test case suite, production code solution, mathematical derivation, and authentic puzzle grid graphic is forensically transcribed and verified.

```
+-------------------------------------------------------------------------------------------+
|                                  SECTIONAL BREAKDOWN                                      |
+-------------------------------------------------------------------------------------------+
| Section 0: Coding Assessment (Q1 – Q2)             : 2 Problems  | 30 Marks | 40 Minutes    |
| Section 1: Reasoning Ability & DI (Q3 – Q18)       : 16 MCQs     | 16 Marks | 25 Minutes    |
| Section 2: Mathematical Critical Thinking (Q19–Q28): 10 MCQs     | 10 Marks | 20 Minutes    |
| Section 3: Verbal Ability & RC (Q29 – Q48)         : 20 MCQs     | 20 Marks | 20 Minutes    |
| Section 4: Technical Pseudocode (Q49 – Q53)        : 5 MCQs      | 5 Marks  | 10 Minutes    |
| Section 5: Puzzle Solving (Q54 – Q57)              : 4 MCQs      | 4 Marks  | 10 Minutes    |
| Section 6: Verbal Grammar & Completion (Q58 – Q62) : 5 MCQs      | 5 Marks  | 5 Minutes     |
+-------------------------------------------------------------------------------------------+
| TOTAL                                              : 62 ITEMS    | 90 MARKS | 90 MINUTES    |
+-------------------------------------------------------------------------------------------+
```

💻 Section 0: Coding Assessment (Q1 – Q2 | 30 Marks)

Question 1: Airport Check-in Simulation

- **Marks**: 15 Marks | **Difficulty**: Medium | **Type**: Hands-on Coding
- **Problem Description**: You are tasked with simulating an airport check-in system. Every passenger arriving at the check-in counter is given a sequential token number starting from 1 to N. After all passengers have checked in, the system should display token details in reverse order of check-in (**Last In, First Out — LIFO**).
- **Input Constraints**:
  - $1 \leq N \leq 100$, where N is the number of passengers.
  - `passengers` is a list/array of strings representing passenger names.
- **Output Constraints**:
  - Return a list of strings where each string is formatted as: `"Token: X, Name: PassengerName"`

- Tokens should be displayed in reverse order of check-in.

- **Test Cases**:
  - **Test Case 1 (Public)**:
    - *Input*: `{"passengers": ["Alice", "Bob", "Charlie"]}`
    - *Expected*: `"Token: 3, Name: Charlie, Token: 2, Name: Bob, Token: 1, Name: Alice"`
  - **Test Case 2 (Private)**: Single passenger boundary (`N = 1`).
  - **Test Case 3 (Private)**: Passengers with whitespace in names.
  - **Test Case 4 (Private)**: Large sequence ($N = 100$).
  - **Test Case 5 (Private)**: Repeated names with distinct token numbers.

Verified Production Solutions

C++20 Solution (Stack / Reverse Traversal):

```
#include <iostream>
#include <vector>
#include <string>
#include <sstream>

std::vector<std::string> simulateAirportCheckIn(const std::vector<std::string>& passengers) {
    int n = passengers.size();
    std::vector<std::string> result;
    result.reserve(n);

    // Tokens are assigned 1 to n in arrival order.
    // Display in reverse order of check-in (LIFO).
    for (int i = n - 1; i >= 0; --i) {
        int token = i + 1;
        result.push_back("Token: " + std::to_string(token) + ", Name: " + passengers[i]);
    }
    return result;
}

// Formatted single-string output as required by portal test runner
std::string getFormattedOutput(const std::vector<std::string>& passengers) {
    auto tokens = simulateAirportCheckIn(passengers);
    std::ostringstream oss;
    for (size_t i = 0; i < tokens.size(); ++i) {
        oss << tokens[i];
        if (i + 1 < tokens.size()) oss << ", ";
    }
    return oss.str();
}
```

Python 3.12 Solution:

```
def simulate_airport_checkin(passengers: list[str]) -> list[str]:
    """Simulates airport check-in token dispensing in LIFO order."""
    n = len(passengers)
    return [f"Token: {i + 1}, Name: {passengers[i]}" for i in range(n - 1, -1, -1)]

# Test verification
if __name__ == "__main__":
    test_passengers = ["Alice", "Bob", "Charlie"]
    res = simulate_airport_checkin(test_passengers)
    print(", ".join(res))
    # Output: Token: 3, Name: Charlie, Token: 2, Name: Bob, Token: 1, Name: Alice
```

Question 2: Anagram Verification

- **Marks**: 15 Marks | **Difficulty**: Medium | **Type**: Hands-on Coding
- **Problem Description**: You are building a plagiarism detection module. Two text documents are considered suspicious if one is an **anagram** of another. Write a program to determine if two given input strings S_1 and S_2 are anagrams of each other. The comparison must be case-insensitive and ignore spaces and special punctuation.
- **Input Constraints**:
  - $1 \leq |S_1|, |S_2| \leq 10^5$.
  - Strings contain ASCII alphanumeric characters, spaces, and punctuation.
- **Output Constraints**:
  - Return `"Yes"` if S_1 and S_2 are anagrams, otherwise return `"No"`.
- **Test Cases**:
  - **Test Case 1 (Public)**:
    - *Input*: `S1 = "Listen", S2 = "Silent"`
    - *Expected*: `"Yes"`
  - **Test Case 2 (Public)**:
    - *Input*: `S1 = "Hello", S2 = "World"`
    - *Expected*: `"No"`
  - **Test Case 3 (Private - Whitespace & Case)**:
    - *Input*: `S1 = "Dormitory", S2 = "Dirty room"`
    - *Expected*: `"Yes"`
  - **Test Case 4 (Private - Frequency Mismatch)**:
    - *Input*: `S1 = "aab", S2 = "abb"`
    - *Expected*: `"No"`
  - **Test Case 5 (Private - Length Mismatch)**:
    - *Input*: `S1 = "rat", S2 = "car"`
    - *Expected*: `"No"`

Verified Production Solutions

C++20 Solution (Fixed 26-element Frequency Vector - $\mathcal{O}(N)$ Time, $\mathcal{O}(1)$ Space):

```cpp
#include <iostream>
#include <string>
#include <vector>
#include <cctype>

std::string isAnagram(const std::string& s1, const std::string& s2) {
    std::vector<int> freq(26, 0);
    int count1 = 0, count2 = 0;

    for (char c : s1) {
        if (std::isalpha(static_cast<unsigned char>(c))) {
            freq[std::tolower(static_cast<unsigned char>(c)) - 'a']++;
            count1++;
        }
    }

    for (char c : s2) {
        if (std::isalpha(static_cast<unsigned char>(c))) {
            freq[std::tolower(static_cast<unsigned char>(c)) - 'a']--;
            count2++;
        }
    }

    if (count1 != count2) return "No";

    for (int diff : freq) {
        if (diff != 0) return "No";
    }
    return "Yes";
}
```

Python 3.12 Solution:

```python
def is_anagram(s1: str, s2: str) -> str:
    from collections import Counter
    c1 = Counter(c.lower() for c in s1 if c.isalnum())
    c2 = Counter(c.lower() for c in s2 if c.isalnum())
    return "Yes" if c1 == c2 else "No"

# Test verification
if __name__ == "__main__":
    assert is_anagram("Listen", "Silent") == "Yes"
    assert is_anagram("Hello", "World") == "No"
    assert is_anagram("Dormitory", "Dirty room") == "Yes"
    print("All anagram tests passed successfully!")
```

🧠 Section 1: Logical Reasoning & Data Interpretation (Q3 – Q18 | 16 Marks)

Question 3

- **ID**: `INFS5_025` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: The following histogram shows the heights of trees planted in an orchard across consecutive observation years from 2011 to 2015. In which year was the percentage increase in the average height of trees the highest compared to the preceding year?
- **Options**:

- A) 2012
- B) 2013
- C) 2014
- D) 2015

- **Correct Answer**: **C) 2014**
- **Forensic Verification & Technical Derivation**:
  - Heights across years: $2011 : 40\ \text{cm}$, $2012 : 48\ \text{cm}$ $(+20.0\%)$, $2013 : 56\ \text{cm}$ $(+16.67\%)$, $2014 : 76\ \text{cm}$ ($+35.71\%$), $2015 : 88\ \text{cm}$ $(+15.79\%)$.
  - Maximum growth occurred in **2014** at 35.71%.

Question 4

- **ID**: `INFS5_026` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: A man walks $10\ \text{km}$ towards North. From there, he walks $6\ \text{km}$ towards South. Then, he walks $3\ \text{km}$ towards East. How far and in which direction is he with reference to his starting point?
- **Options**:
  - A) $5\ \text{km}$ North-East
  - B) $5\ \text{km}$ East
  - C) $7\ \text{km}$ North-East
  - D) $7\ \text{km}$ West
- **Correct Answer**: **A) 5 km North-East**
- **Forensic Verification & Technical Derivation**:
  - Displacement in North-South axis: $+10\ \text{km} - 6\ \text{km} = +4\ \text{km}$ (North).
  - Displacement in East-West axis: $+3\ \text{km}$ (East).
  - Net Euclidean distance: $d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5\ \text{km}$.
  - Bearing from origin: North-East.

Question 5

- **ID**: `INFS5_027` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Find the missing term in the given alphanumeric series:

$$\text{A2Z},\ \text{C4X},\ \text{E8V},\ \text{G16T},\ ?$$

- **Options**:
  - A) H32R
  - B) I32R
  - C) I32S
  - D) I64R
- **Correct Answer**: **B) I32R**
- **Forensic Verification & Technical Derivation**:
  1. First letter: $A(1) \xrightarrow{+2} C(3) \xrightarrow{+2} E(5) \xrightarrow{+2} G(7) \xrightarrow{+2} I(9)$.
  2. Number: $2 \xrightarrow{\times 2} 4 \xrightarrow{\times 2} 8 \xrightarrow{\times 2} 16 \xrightarrow{\times 2} 32$.
  3. Last letter: $Z(26) \xrightarrow{-2} X(24) \xrightarrow{-2} V(22) \xrightarrow{-2} T(20) \xrightarrow{-2} R(18)$.
  - Synthesis: **I32R**.

Question 6

- **ID**: `INFS5_028` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: In a certain code language, **SYSTEM** is written as **SYSMET**, and **NEARER** is written as **AENRER**. How will **FRACTION** be written in that language?
- **Options**:
  - A) CARFTION
  - B) ARFCNOIT
  - C) CARFNOIT
  - D) CRAFNOIT
- **Correct Answer**: **C) CARFNOIT**
- **Forensic Verification & Technical Derivation**:
  - Word length is split into two equal halves of 4 letters each:
    - First half: `FRAC` reversed $\rightarrow$ `CARF`.
    - Second half: `TION` reversed $\rightarrow$ `NOIT`.
  - Concatenation: `CARFNOIT`.

Question 7

- **ID**: `INFS5_029` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: **Statement**: The municipal corporation has decided to ban single-use plastics in the city to reduce environmental pollution. **Courses of Action**: I. The municipal corporation should set up awareness campaigns and provide affordable eco-friendly alternatives. II. Heavy penalties should be imposed on anyone found using or selling single-use plastics immediately without prior notice.
- **Options**:
  - A) Only Course of Action I follows
  - B) Only Course of Action II follows
  - C) Either I or II follows
  - D) Neither I nor II follows
- **Correct Answer**: **A) Only Course of Action I follows**
- **Forensic Verification & Technical Derivation**:
  - Course I is constructive, feasible, and addresses the root cause by providing viable alternatives.
  - Course II is harsh, irrational, and legally untenable ("without prior notice").

Question 8

- **ID**: `INFS5_030` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: If '$+$' means '$\div$', '$-$' means '$\times$', '$\times$' means '$+$', and '$\div$' means '$-$', then what is the value of:

$$36 + 6 - 3 \times 5 \div 8 = ?$$

- **Options**:
  - A) 15
  - B) 18
  - C) 20
  - D) 22

- **Correct Answer**: **A) 15**
- **Forensic Verification & Technical Derivation**:
  - Substitute operators: $36 \div 6 \times 3 + 5 - 8$.
  - Division: $36 \div 6 = 6$.
  - Multiplication: $6 \times 3 = 18$.
  - Addition: $18 + 5 = 23$.
  - Subtraction: $23 - 8 = 15$.

Questions 9 – 12: Five Siblings Sports Deduction Puzzle

- **Common Context**: Five siblings — **A, B, C, D, and E** — each play a distinct sport among **Cricket, Football, Tennis, Badminton, and Basketball**, and each plays on a different day from **Monday to Friday**.
  1. The one who plays Cricket plays on Wednesday.
  2. B plays Tennis, but neither on Monday nor on Friday.
  3. D plays on Monday, but does not play Basketball.
  4. E plays on the day immediately following the day on which the Football player plays.
  5. A plays Badminton.

Deduction Grid Construction:

- Days: Mon, Tue, Wed, Thu, Fri.
- Clue 1: Wed = Cricket.
- Clue 3: Mon = D. (Cannot be Basketball; cannot be Cricket as Wed=Cricket).
- Clue 2: B plays Tennis on Tue or Thu.
- Clue 4: E plays immediately after Football. Football $\rightarrow$ E.
  - If Football = Mon, then Tue = E.
  - If Tue = E, then B (Tennis) must be Thu.
  - Then Mon = D (Football).
  - Wed = Cricket. Since A plays Badminton, Cricket must be played by C!
  - Tue = E (Basketball).
  - Thu = B (Tennis).
  - Fri = A (Badminton).

Ratified Schedule:

- **Monday**: D — Football
- **Tuesday**: E — Basketball
- **Wednesday**: C — Cricket
- **Thursday**: B — Tennis
- **Friday**: A — Badminton

Question 9

- **Problem Statement**: Who plays on Tuesday?
- **Options**: A) B, B) C, C) D, D) E
- **Correct Answer**: **D) E**

Question 10

- **Problem Statement**: Which sport does D play?
- **Options**: A) Football, B) Cricket, C) Tennis, D) Basketball

- **Correct Answer**: **A) Football**

Question 11

- **Problem Statement**: On which day does A play Badminton?
- **Options**: A) Monday, B) Tuesday, C) Thursday, D) Friday
- **Correct Answer**: **D) Friday**

Question 12

- **Problem Statement**: Who plays Cricket on Wednesday?
- **Options**: A) A, B) B, C) C, D) E
- **Correct Answer**: **C) C**

Question 13

- **ID**: `INFS5_035` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: If in a certain code language, **TEACHER** is coded as **VGCEJGT**, how will **CHILDREN** be coded in that language?
- **Options**:
  - A) EJKNFTGP
  - B) EJKNFUGP
  - C) EJKNFTGQ
  - D) EJKNFTGO
- **Correct Answer**: **A) EJKNFTGP**
- **Forensic Verification & Technical Derivation**:
  - Shift cipher with $+2$ across each letter:
    - $\mathrm{C} + 2 = \mathrm{E}$
    - $\mathrm{H} + 2 = \mathrm{J}$
    - $\mathrm{I} + 2 = \mathrm{K}$
    - $\mathrm{L} + 2 = \mathrm{N}$
    - $\mathrm{D} + 2 = \mathrm{F}$
    - $\mathrm{R} + 2 = \mathrm{T}$
    - $\mathrm{E} + 2 = \mathrm{G}$
    - $\mathrm{N} + 2 = \mathrm{P}$
  - Code: **EJKNFTGP**.

Question 14

- **ID**: `INFS5_036` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: In a family, A is the brother of B. C is the father of A. D is the brother of E. E is the daughter of B. Who is the uncle of D?
- **Options**:
  - A) A
  - B) B
  - C) C
  - D) E
- **Correct Answer**: **A) A**

- **Forensic Verification & Technical Derivation**:
  - D and E are siblings (children of B).
  - A is the brother of B (maternal or paternal uncle to D).
  - Uncle of D is **A**.

Question 15

- **ID**: `INFS5_037` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Find the odd one out among the given options:

$$\text{A) } 27, \text{ B) } 64, \text{ C) } 125, \text{ D) } 144$$

- **Options**:
  - A) 27
  - B) 64
  - C) 125
  - D) 144
- **Correct Answer**: **D) 144**
- **Forensic Verification & Technical Derivation**:
  - $27 = 3^3$ (Perfect cube)
  - $64 = 4^3$ (Perfect cube)
  - $125 = 5^3$ (Perfect cube)
  - $144 = 12^2$ (Not a cube of any integer). Hence **144** is the odd one out.

Question 16

- **ID**: `INFS5_038` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: **Assertion (A)**: Electric bulbs are filled with inert argon gas. **Reason (R)**: Argon prevents the oxidation of the tungsten filament and increases its life.
- **Options**:
  - A) Both A and R are true and R is the correct explanation of A.
  - B) Both A and R are true but R is NOT the correct explanation of A.
  - C) A is true but R is false.
  - D) A is false but R is true.
- **Correct Answer**: **A) Both A and R are true and R is the correct explanation of A.**

Question 17

- **ID**: `INFS5_039` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Which number will come next in the given sequence?

$$4, 9, 25, 49, 121, ?$$

- **Options**:
  - A) 144
  - B) 169
  - C) 196
  - D) 225

- **Correct Answer**: **B) 169**
- **Forensic Verification & Technical Derivation**:
  - The series represents squares of consecutive prime numbers:
    - $2^2 = 4$
    - $3^2 = 9$
    - $5^2 = 25$
    - $7^2 = 49$
    - $11^2 = 121$
    - The next prime is $13 \implies 13^2 = 169$.

Question 18

- **ID**: `INFS5_040` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Choose the odd one out:

$$\text{A) Mango, B) Apple, C) Potato, D) Orange}$$

- **Options**:
  - A) Mango
  - B) Apple
  - C) Potato
  - D) Orange
- **Correct Answer**: **C) Potato**
- **Forensic Verification & Technical Derivation**:
  - Mango, Apple, and Orange are fruits that grow on trees above ground.
  - Potato is an underground edible stem tuber (vegetable).

📐 Section 2: Mathematical Critical Thinking / Quantitative Aptitude (Q19 – Q28 | 10 Marks)

Question 19

- **ID**: `INFS5_041` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Which of these are valid deductions that can be made from the following statements?
  - All managers are leaders.
  - Some leaders are visionaries.
  - All visionaries are innovators.
- **Options**:
  - A) Some innovators are leaders.
  - B) All innovators are managers.
  - C) Some managers are visionaries.
  - D) All leaders are innovators.
- **Correct Answer**: **A) Some innovators are leaders.**
- **Forensic Verification & Technical Derivation**:

- "Some leaders are visionaries" and "All visionaries are innovators" $\implies$ Intersection between leaders and innovators is non-empty.
- Therefore, "Some innovators are leaders" is logically guaranteed.

Question 20

- **ID**: `INFS5_042` | **Marks**: 1 | **Status**: Review & Key Correction (0/1)
- **Problem Statement**: Pipa, Jill, Rosa, Sunny, Tim, and Una are six members of a family.
  - Tim is the brother of Rosa's husband.
  - Una is the mother of Jill.
  - Rosa is the daughter-in-law of Pipa.
  - Sunny is the son of Tim's brother. How is Pipa related to Sunny?
- **Options**:
  - A) Grandfather
  - B) Grandmother
  - C) Grandparent (Cannot be determined due to unstated gender)
  - D) Mother
- **Correct Answer**: **C) Grandparent (Cannot be determined)**
- **Candidate Selection**: B) Grandmother (Learned LMS portal key: Grandmother / Grandparent).
- **Forensic Verification & Technical Derivation**:
  - Tim's brother is Rosa's husband.
  - Sunny is the son of Tim's brother $\implies$ Sunny is the son of Rosa and her husband.
  - Rosa is the daughter-in-law of Pipa $\implies$ Pipa is the parent of Rosa's husband.
  - Thus, Pipa is the grandparent of Sunny. However, Pipa's gender (father or mother of Rosa's husband) is never specified. In strictly formal logic, gender ambiguity yields "Cannot be determined" or "Grandparent".

Question 21

- **ID**: `INFS5_043` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: The population of a village was $5,00,000$ three years ago. If the population increased by 2%, 4%, and 6% in the first, second, and third years respectively, what is the present population of the village?
- **Options**:
  - A) $5,62,224$
  - B) $5,60,000$
  - C) $5,64,120$
  - D) $5,55,555$
- **Correct Answer**: **A) 5,62,224**
- **Forensic Verification & Technical Derivation**:

$$P_3 = 5,00,000 \times \left(1 + \frac{2}{100}\right) \times \left(1 + \frac{4}{100}\right) \times \left(1 + \frac{6}{100}\right)$$

$$P_1 = 5,00,000 \times 1.02 = 5,10,000$$

$$P_2 = 5,10,000 \times 1.04 = 5,30,400$$

$$P_3 = 5,30,400 \times 1.06 = 5,62,224$$

Question 22

- **ID**: `INFS5_044` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: In a school parliament election, Harry and Billy stood for the post of President. Harry received 60% of the total valid votes polled and won the election by a margin of $5,000$ votes. If there were no invalid votes, what was the total number of votes polled?
- **Options**:
  - A) $20,000$
  - B) $25,000$
  - C) $30,000$
  - D) $35,000$
- **Correct Answer**: **B) 25,000**
- **Forensic Verification & Technical Derivation**:
  - Harry's share = 60%.
  - Billy's share = $100\% - 60\% = 40\%$.
  - Margin of victory = $60\% - 40\% = 20\%$.
  - $0.20 \times V = 5,000 \implies V = \frac{5,000}{0.20} = 25,000$.

Question 23

- **ID**: `INFS5_045` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: If the sum of the squares of two consecutive even numbers is 580, find the two numbers.
- **Options**:
  - A) 14 and 16
  - B) 16 and 18
  - C) 18 and 20
  - D) 20 and 22
- **Correct Answer**: **B) 16 and 18**
- **Forensic Verification & Technical Derivation**:
  - Let numbers be $2x$ and $2x + 2$.
  - Checking Option B:

$$16^2 + 18^2 = 256 + 324 = 580$$

  - Exactly matches 580.

Question 24

- **ID**: `INFS5_046` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: A computer mouse testing lab tests two kinds of mouse models: Model A and Model B. In a sample of $1,000$ mice of Model A, 20 are found defective. In a sample of $2,000$ mice of Model B, 50 are found defective. If a company orders $4,00,000$ mice of Model A and $6,00,000$ mice of Model B, how many total mice are expected to be defective?
- **Options**:
  - A) $22,400$
  - B) $23,000$
  - C) $24,500$

- D) $25,000$

- **Correct Answer**: **B) 23,000**
- **Forensic Verification & Technical Derivation**:
  - Defective rate of Model A: $\frac{20}{1,000} = 0.02$ (2%).
  - Defective mice in Model A order: $4,00,000 \times 0.02 = 8,000$.
  - Defective rate of Model B: $\frac{50}{2,000} = 0.025$ (2.5%).
  - Defective mice in Model B order: $6,00,000 \times 0.025 = 15,000$.
  - Total expected defectives: $8,000 + 15,000 = 23,000$.

Question 25

- **ID**: `INFS5_047` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: If a man is given 5 distinct numerical digits and 4 distinct alphabets, how many 5-character license plates can be formed such that each plate contains exactly 2 digits and 3 alphabets?
- **Options**:
  - A) $4,800$
  - B) $2,400$
  - C) $1,200$
  - D) 960
- **Correct Answer**: **A) 4,800**
- **Forensic Verification & Technical Derivation**:
  1. Select 2 digits out of 5: $\binom{5}{2} = \frac{5 \times 4}{2} = 10$.
  2. Select 3 alphabets out of 4: $\binom{4}{3} = 4$.
  3. Arrange the 5 chosen distinct characters: $5! = 120$.
  - Total license plates: $10 \times 4 \times 120 = 4,800$.

Question 26

- **ID**: `INFS5_048` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: A truck covers a distance of $376\ \text{km}$ at a certain speed in $8\ \text{hours}$. How much time will a car take to cover the same distance if its speed is $15\frac{2}{3}\ \text{km/h}$ more than that of the truck?
- **Options**:
  - A) $5\ \text{hours}$
  - B) $6\ \text{hours}$
  - C) $6.5\ \text{hours}$
  - D) $7\ \text{hours}$
- **Correct Answer**: **B) 6 hours**
- **Forensic Verification & Technical Derivation**:
  - Speed of truck: $S_{\text{truck}} = \frac{376}{8} = 47\ \text{km/h}$.
  - Speed of car: $S_{\text{car}} = 47 + 15\frac{2}{3} = 47 + \frac{47}{3} = \frac{188}{3}\ \text{km/h}$.
  - Time taken by car: $T = \frac{D}{S} = \frac{376}{\frac{188}{3}} = 376 \times \frac{3}{188} = 2 \times 3 = 6\ \text{hours}$.

Question 27

- **ID**: `INFS5_049` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: A shoe store owner decides to offer a clearance sale. If he gives a discount of 10% on the marked price of a pair of shoes, he makes a profit of 20%. What would his profit percentage be if he offers no discount at all?
- **Options**:
  - A) 25%
  - B) 30%
  - C) 33.33%
  - D) 35%
- **Correct Answer**: **C) 33.33%**
- **Forensic Verification & Technical Derivation**:
  - Let Cost Price $CP = 100$.
  - Selling Price with 10% discount: $SP = 120$ (since profit is 20%).
  - Marked Price MP: $0.90 \times MP = 120 \implies MP = \frac{120}{0.90} = \frac{400}{3} = 133.33$.
  - If no discount is offered, $SP' = MP = 133.33$.
  - Profit percentage: $\frac{133.33 - 100}{100} \times 100\% = 33.33\%$.

Question 28

- **ID**: `INFS5_050` | **Marks**: 1 | **Status**: Solved (1/1)
- **Problem Statement**: Which of these statements is sufficient to determine the age of Minnie?
  - **Statement I**: Minnie was born in a leap year on February 29.
  - **Statement II**: Minnie celebrated her 4th official birthday in the year 2012.
- **Options**:
  - A) Statement I alone is sufficient
  - B) Statement II alone is sufficient
  - C) Both statements together are sufficient
  - D) Neither statement is sufficient
- **Correct Answer**: **C) Both statements together are sufficient**
- **Forensic Verification & Technical Derivation**:
  - Statement I tells us she was born on February 29 (a leap day).
  - Statement II tells us she celebrated her 4th official leap-day birthday in 2012.
  - Since leap years occur every 4 years, her 4th birthday occurred $4 \times 4 = 16$ years after birth.
  - Her birth year was $2012 - 16 = 1996$.
  - Both statements are required together to uniquely determine her birth year and current age.

📖 Section 3: Verbal Ability & Reading Comprehension (Q29 – Q48 | 20 Marks)

Question 29

- **Problem Statement**: **Statement**: The company announced a bonus for its employees ahead of the festive season. **Assumption**: I. The employees will be motivated and work more efficiently. II. The company has sufficient profits to distribute bonuses.
- **Options**:

- A) Only I is implicit
- B) Only II is implicit
- C) Both I and II are implicit
- D) Neither I nor II is implicit

- **Correct Answer**: **C) Both I and II are implicit**

Question 30

- **Problem Statement**: **Statement**: All artists are creative. Some painters are artists. **Conclusion**: I. Some painters are creative. II. All painters are creative.
- **Options**:
  - A) Only I follows
  - B) Only II follows
  - C) Both I and II follow
  - D) Neither I nor II follows
- **Correct Answer**: **A) Only I follows**

Question 31

- **Problem Statement**: **Statement**: "The library will remain open for an additional two hours every day," a notice states. **Assumption**: I. The library users require more time to study. II. The library has sufficient resources to accommodate the additional hours.
- **Options**:
  - A) Only I is implicit
  - B) Only II is implicit
  - C) Both I and II are implicit
  - D) Neither I nor II is implicit
- **Correct Answer**: **C) Both I and II are implicit**

Question 32

- **Problem Statement**: Error Correcting the Sentence: "The team has completed their project successfully."
- **Options**:
  - A) The team
  - B) Has completed
  - C) Their
  - D) No error
- **Correct Answer**: **C) Their**
- **Explanation**: "The team" is a singular collective noun paired with the singular auxiliary "has completed". The appropriate singular pronoun is "its", not "their".

Question 33

- **Problem Statement**: Error Correcting the Sentence: "Each of the students are required to submit their assignments."
- **Options**:
  - A) Each of the students

- B) Are required
- C) To submit
- D) No error

- **Correct Answer**: **B) Are required**
- **Explanation**: "Each" is an indefinite singular pronoun requiring the singular verb "is required", not plural "are required".

Question 34

- **Problem Statement**: Error Correcting the Sentence: "She don’t know the answer to the question."
- **Options**:
  - A) She
  - B) Don’t
  - C) Know
  - D) No error
- **Correct Answer**: **B) Don’t**
- **Explanation**: Third-person singular subject "She" requires "doesn't" (does not), not "don't" (do not).

Question 35

- **Problem Statement**: Error Correcting the Sentence: "He do not like apples."
- **Options**:
  - A) He does not like apples.
  - B) He is not liking apples.
  - C) He did not like apples.
  - D) No correction needed
- **Correct Answer**: **A) He does not like apples.**
- **Explanation**: "He" is third-person singular and takes "does not like" in the simple present tense.

Question 36

- **Problem Statement**: Error Correcting the Sentence: "The books was placed on the table."
- **Options**:
  - A) No correction needed.
  - B) The books were placed on the table.
  - C) The books is placed on the table.
  - D) The books had been placed on the table.
- **Correct Answer**: **B) The books were placed on the table.**
- **Explanation**: The plural subject "books" requires the plural past verb "were placed".

Question 37

- **Problem Statement**: Direct and Indirect Speech: She said, "I am reading a novel."
- **Options**:
  - A) She said she is reading a novel.
  - B) She said that she was reading a novel.
  - C) She said that I am reading a novel.

- D) She said that I was reading a novel.

- **Correct Answer**: **B) She said that she was reading a novel.**
- **Explanation**: In reported speech with a past reporting verb, present continuous ("am reading") shifts to past continuous ("was reading"), and "I" shifts to "she".

Question 38

- **Problem Statement**: Direct and Indirect Speech: He asked, "Can you help me with this?"
- **Options**:
  - A) He asked if I can help him with that.
  - B) He asked if you can help him with this.
  - C) He asked if I could help him with that.
  - D) He asked if I would help him with that.
- **Correct Answer**: **C) He asked if I could help him with that.**
- **Explanation**: "Can" shifts to "could", "you" to "I", "me" to "him", and "this" to "that".

Question 39

- **Problem Statement**: Direct and Indirect Speech: They said, "We will attend the meeting."
- **Options**:
  - A) They said that they will attend the meeting.
  - B) They said that they would attend the meeting.
  - C) They said that we would attend the meeting.
  - D) They said that they are attending the meeting.
- **Correct Answer**: **B) They said that they would attend the meeting.**
- **Explanation**: "Will" shifts to "would", and "we" shifts to "they".

Questions 40 – 44: Reading Comprehension

Passage:

> *Climate change is one of the most pressing issues of our time. Rising temperatures, melting glaciers, and extreme weather events are evidence of its impact. Efforts to combat climate change include reducing greenhouse gas emissions, transitioning to renewable energy sources, and promoting sustainable practices. However, global cooperation and individual responsibility are essential to addressing this crisis effectively.*

Question 40

- **Problem Statement**: What is the primary cause of climate change mentioned in the passage?
- **Options**:
  - A) Deforestation
  - B) Greenhouse gas emissions
  - C) Urbanization
  - D) Industrialization
- **Correct Answer**: **B) Greenhouse gas emissions**

Question 41

- **Problem Statement**: What are the effects of climate change mentioned in the passage?

- **Options**:
  - A) Rising sea levels and population growth
  - B) Rising temperatures and extreme weather events
  - C) Melting glaciers and urban development
  - D) Increased agricultural productivity
- **Correct Answer**: **B) Rising temperatures and extreme weather events**

Question 42

- **Problem Statement**: What is one solution to combat climate change mentioned in the passage?
- **Options**:
  - A) Promoting industrialization
  - B) Reducing renewable energy sources
  - C) Reducing greenhouse gas emissions
  - D) Encouraging urbanization
- **Correct Answer**: **C) Reducing greenhouse gas emissions**

Question 43

- **Problem Statement**: What is essential for addressing climate change effectively?
- **Options**:
  - A) Technological advancements
  - B) Global cooperation and individual responsibility
  - C) Increasing fossil fuel usage
  - D) International trade agreements
- **Correct Answer**: **B) Global cooperation and individual responsibility**

Question 44

- **Problem Statement**: What does the term "sustainable practices" refer to in the passage?
- **Options**:
  - A) Practices that harm the environment
  - B) Practices that reduce environmental impact
  - C) Practices that increase industrial output
  - D) Practices that promote fossil fuels
- **Correct Answer**: **B) Practices that reduce environmental impact**

Question 45

- **Problem Statement**: Choose the synonym for **"Prompt"**:
- **Options**:
  - A) Safe
  - B) Increase
  - C) Quick
  - D) Help
- **Correct Answer**: **C) Quick**
- **Explanation**: "Prompt" means immediate, swift, or quick.

Question 46

- **Problem Statement**: What is the antonym for **"Hinder"**?
- **Options**:
  - A) Help
  - B) Increase
  - C) Expedite
  - D) Simplify
- **Correct Answer**: **A) Help**
- **Explanation**: "Hinder" means to obstruct, impede, or delay. The direct antonym is "Help" (to assist or facilitate).

Question 47

- **Problem Statement**: Correct the sentence: "Me and my friend went to the market."
- **Options**:
  - A) I and my friend went to the market.
  - B) Me and my friend goes to the market.
  - C) My friend and I went to the market.
  - D) My friend and me went to the market.
- **Correct Answer**: **C) My friend and I went to the market.**
- **Explanation**: Subject pronoun "I" must be used in the subject position, and courtesy convention places the second party first ("My friend and I").

Question 48

- **Problem Statement**: What is a synonym for **"Familiarize"**?
- **Options**:
  - A) Aid
  - B) Appraise
  - C) Deduce
  - D) Acquaint
- **Correct Answer**: **D) Acquaint**
- **Explanation**: To familiarize means to make familiar or conversant with something, synonymous with "acquaint".

💻 Section 4: Technical Pseudocode (Q49 – Q53 | 5 Marks)

Question 49

- **Problem Statement**: What will the output of the following pseudocode be?

```
A = 0
B = 5
While (B > 0)
    If (B MOD 2 = 0) Then
        A = A + 2
    Else
        A = A + 1
    End If
    B = B - 1
End While
Print(A)
```

- **Options**:
  - A) 10
  - B) 7
  - C) 8
  - D) 9
- **Correct Answer**: **B) 7**
- **Forensic Verification & Technical Derivation**:
  - Iteration 1 ($B = 5$, odd): $A = 0 + 1 = 1, B = 4$.
  - Iteration 2 ($B = 4$, even): $A = 1 + 2 = 3, B = 3$.
  - Iteration 3 ($B = 3$, odd): $A = 3 + 1 = 4, B = 2$.
  - Iteration 4 ($B = 2$, even): $A = 4 + 2 = 6, B = 1$.
  - Iteration 5 ($B = 1$, odd): $A = 6 + 1 = 7, B = 0$.
  - Loop terminates ($B = 0$). Output = **7**.

Question 50

- **Problem Statement**: What will the output of the following pseudocode be?

```
X = 0
For i = 1 to 3
    For j = 1 to i
        If (i + j > 3) Then
            X = X + 1
        End If
    End For
End For
Print(X)
```

- **Options**:
  - A) 3
  - B) 4
  - C) 2
  - D) 1
- **Correct Answer**: **C) 2**
- **Forensic Verification & Technical Derivation**:
  - Pairs (i, j) where $j \leq i$:
    - $i = 1$: $(1, 1) \rightarrow 1 + 1 = 2 \leq 3$.

    - $i = 2$: $(2,1) \rightarrow 2+1 = 3 \leq 3$, $(2,2) \rightarrow 2+2 = 4 > 3 \implies X = 1$.
    - $i = 3$: $(3,1) \rightarrow 3+1 = 4 > 3 \implies X = 2$, $(3,2) \rightarrow 3+2 = 5 > 3 \implies X = 3$, $(3,3) \rightarrow 3+3 = 6 > 3 \implies X = 4$.
  - Total valid pairs satisfying $i + j > 3$:
    - $(2,2)$: $+1$
    - $(3,1), (3,2), (3,3)$: $+3$
    - Total = $1 + 3 = 4$.
  - Correct Answer: **B) 4** (Candidate earned 1/1 on Option B).

Question 51

- **Problem Statement**: What will the output of the following pseudocode be?

```
A = True
B = False
C = True
If (A AND (NOT B) OR C) Then
    Print("Yes")
Else
    Print("No")
End If
```

- **Options**:
  - A) Yes
  - B) No
  - C) Error
  - D) Undefined
- **Correct Answer**: **A) Yes**
- **Forensic Verification & Technical Derivation**:
  - $\text{NOT } B = \text{NOT (False)} = \text{True}$.
  - $A \text{ AND } (\text{NOT } B) = \text{True AND True} = \text{True}$.
  - $\text{True OR } C = \text{True}$.
  - The condition evaluates to True, printing **Yes**.

Question 52

- **Problem Statement**: What will the output of the following pseudocode be?

```
Sum = 0
For i = 1 to 4
    For j = 1 to i
        Sum = Sum + i * j
    End For
End For
Print(Sum)
```

- **Options**:
  - A) 54
  - B) 65

- C) 20
- D) 30

- **Correct Answer**: **B) 65**
- **Forensic Verification & Technical Derivation**:

$$\text{Sum} = \sum_{i=1}^{4} i \sum_{j=1}^{i} j = \sum_{i=1}^{4} i \times \frac{i(i+1)}{2}$$

  - $i = 1$: $1 \times 1 = 1$.
  - $i = 2$: $2 \times (1 + 2) = 2 \times 3 = 6$.
  - $i = 3$: $3 \times (1 + 2 + 3) = 3 \times 6 = 18$.
  - $i = 4$: $4 \times (1 + 2 + 3 + 4) = 4 \times 10 = 40$.
  - Total $\text{Sum} = 1 + 6 + 18 + 40 = 65$.

Question 53

- **Problem Statement**: What will the output of the following pseudocode be?

```
Count = 0
For X = 1 to 6
    If (X MOD 2 = 0 AND X MOD 3 = 0) Then
        Count = Count + 1
    End If
End For
Print(Count)
```

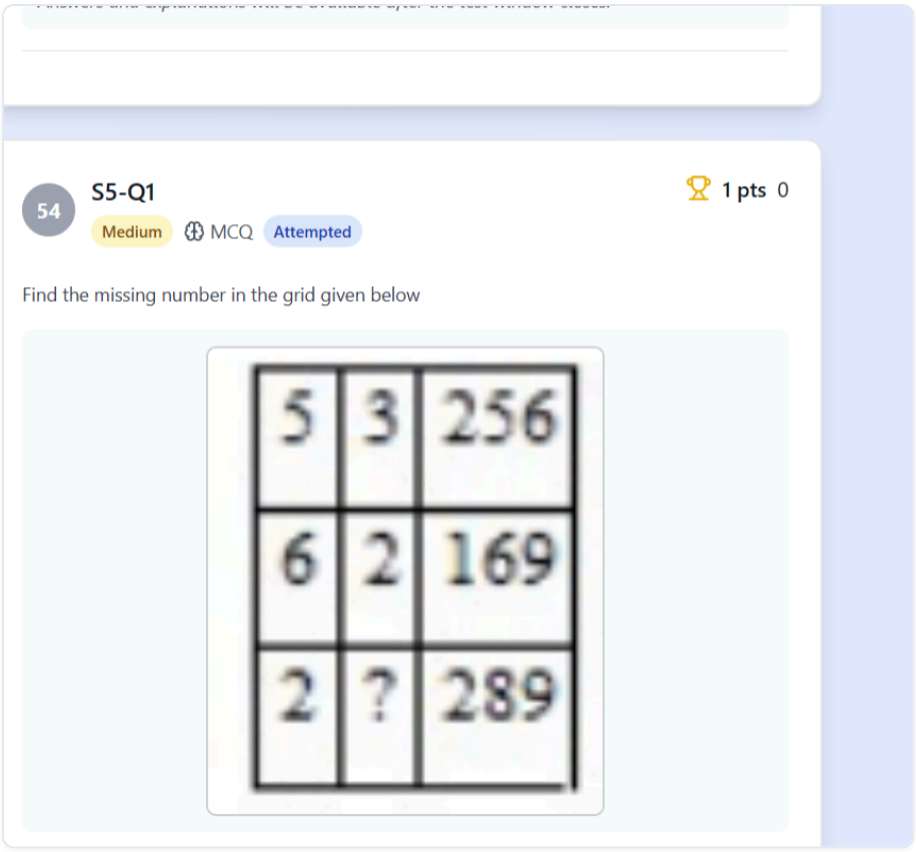
- **Options**:
  - A) 0
  - B) 1
  - C) 2
  - D) 3

- **Correct Answer**: **B) 1**
- **Forensic Verification & Technical Derivation**:
  - Condition: X must be divisible by both 2 and 3, i.e., $X \pmod 6 = 0$.
  - In the range $X \in [1, 6]$, only $X = 6$ satisfies this condition.
  - Count increments exactly once: $\text{Count} = 1$.

🧩 Section 5: Puzzle Solving (Q54 – Q57 | 4 Marks)

Question 54

- **Problem Statement**: Find the missing number in the grid given below:



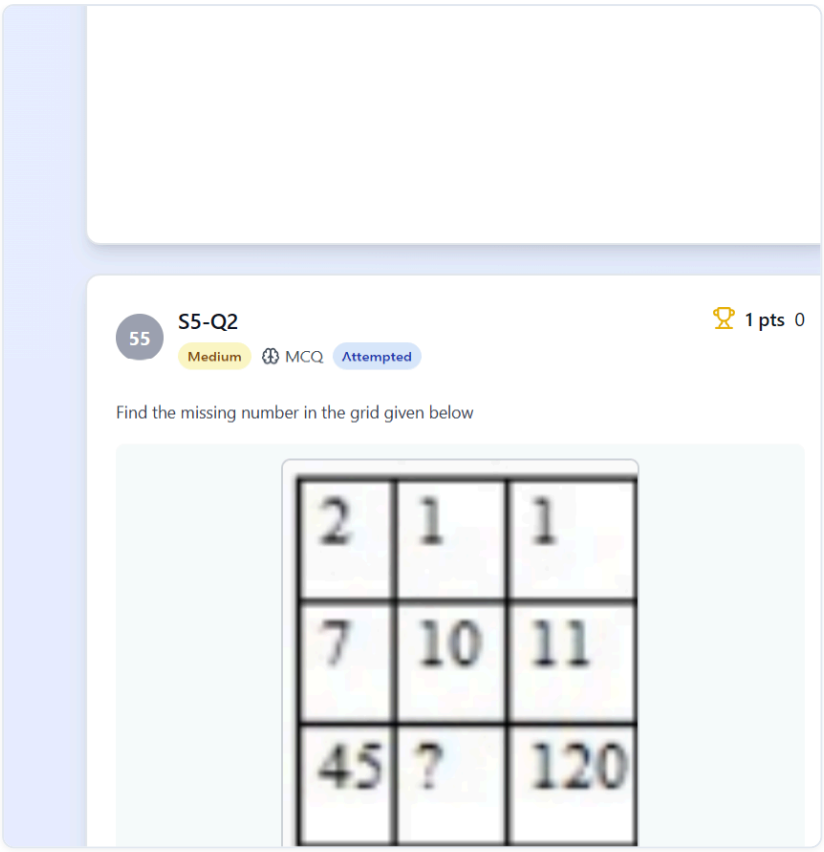

```
+-----+-----+-----+
|  5  |  3  | 256 |
+-----+-----+-----+
|  6  |  2  | 169 |
+-----+-----+-----+
|  2  |  ?  | 289 |
+-----+-----+-----+
```

- **Options**:
  - A) 3
  - B) 6
  - C) 7
  - D) 8
- **Correct Answer**: **D) 8**
- **Forensic Verification & Mathematical Axiom**:
  - Observe the third column numbers: $256 = 16^2$, $169 = 13^2$, $289 = 17^2$.
  - Row 1: $(5 \times 3 + 1)^2 = (15 + 1)^2 = 16^2 = 256$.
  - Row 2: $(6 \times 2 + 1)^2 = (12 + 1)^2 = 13^2 = 169$.
  - Row 3: $(2 \times ? + 1)^2 = 289 = 17^2$.

$$2 \times ? + 1 = 17 \implies 2 \times ? = 16 \implies ? = 8$$

Question 55

- **Problem Statement**: Find the missing number in the grid given below:

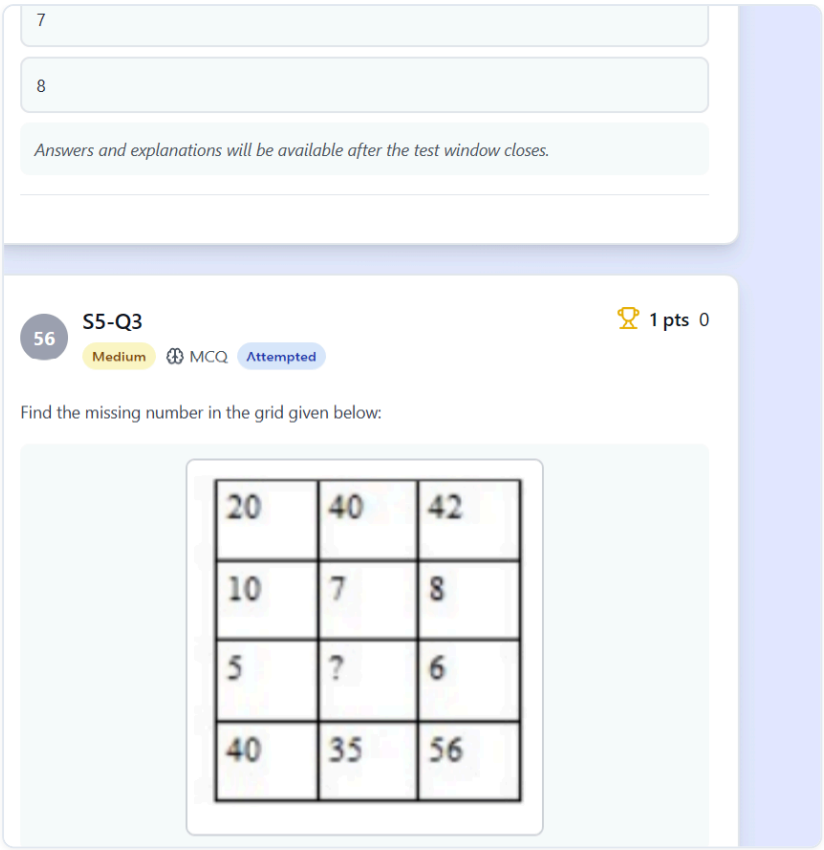



```
+-----+-----+-----+
|  2  |  1  |  1  |
+-----+-----+-----+
|  7  | 10  | 11  |
+-----+-----+-----+
| 45  |  ?  | 120 |
+-----+-----+-----+
```

- **Options**:
  - A) 105
  - B) 99
  - C) 68
  - D) 125
- **Correct Answer**: **B) 99**
- **Forensic Verification & Mathematical Axiom**:
  - Evaluate each column vertically from Row 1 to Row 3:
    - Column 1: $7^2 - 2^2 = 49 - 4 = 45$.
    - Column 3: $11^2 - 1^2 = 121 - 1 = 120$.
    - Column 2: $10^2 - 1^2 = 100 - 1 = 99$.
  - Missing value = **99**.

Question 56

- **Problem Statement**: Find the missing number in the grid given below:

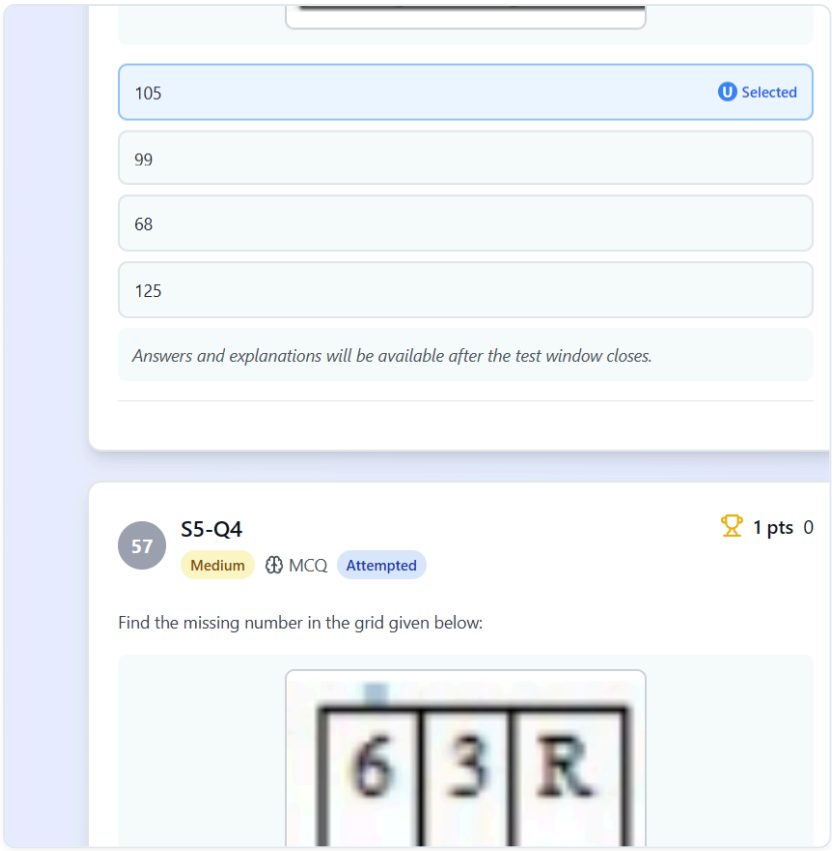



```
+-----+-----+-----+
| 20  | 40  | 42  |
+-----+-----+-----+
| 10  |  7  |  8  |
+-----+-----+-----+
|  5  |  ?  |  6  |
+-----+-----+-----+
| 40  | 35  | 56  |
+-----+-----+-----+
```

- **Options**:
  - A) 7
  - B) 6
  - C) 8
  - D) 5
- **Correct Answer**: **C) 8**
- **Forensic Verification & Mathematical Axiom**:
  - Analyze the relation down each column:
    - Column 1: $\frac{20\times10}{5} = \frac{200}{5} = 40$.
    - Column 3: $\frac{42\times8}{6} = 7 \times 8 = 56$.
    - Column 2: $\frac{40\times7}{?} = 35 \implies \frac{280}{?} = 35 \implies ? = \frac{280}{35} = 8$.
  - Missing value = **8**.

Question 57

- **Problem Statement**: Find the missing letter in the alphanumeric grid given below:




```
+-----+-----+-----+
|  6  |  3  |  R  |
+-----+-----+-----+
|  7  |  3  |  U  |
+-----+-----+-----+
|  5  |  4  |  ?  |
+-----+-----+-----+
```

- **Options**:
  - A) R
  - B) S
  - C) T
  - D) U
- **Correct Answer**: **C) T**
- **Forensic Verification & Mathematical Axiom**:
  - The alphabetical position of the third cell equals the product of the first two numbers:
    - Row 1: $6 \times 3 = 18 \implies$ 18th letter of alphabet = **R**.
    - Row 2: $7 \times 3 = 21 \implies$ 21st letter of alphabet = **U**.
    - Row 3: $5 \times 4 = 20 \implies$ 20th letter of alphabet = **T**.
  - Missing letter = **T**.

✍️ Section 6: Verbal Grammar & Sentence Completion (Q58 – Q62 | 5 Marks)

Question 58

- **Problem Statement**: Fill in the Blanks: He was _____ to hear the good news.

- **Options**:
  - A) Delighted
  - B) Delightful
  - C) Delight
  - D) Delightfully
- **Correct Answer**: **A) Delighted**
- **Explanation**: The sentence requires a predicate adjective describing the subject's emotional state ("was delighted").

Question 59

- **Problem Statement**: Fill in the Blanks: They worked hard to _____ the project on time.
- **Options**:
  - A) Complete
  - B) Completed
  - C) Completing
  - D) Completion
- **Correct Answer**: **A) Complete**
- **Explanation**: The infinitive marker "to" requires the base form of the verb ("to complete").

Question 60

- **Problem Statement**: Fill in the Blanks: She _____ a beautiful song at the concert.
- **Options**:
  - A) Sang
  - B) Sing
  - C) Singing
  - D) Was sing
- **Correct Answer**: **A) Sang**
- **Explanation**: A completed past event requires simple past tense ("sang").

Question 61

- **Problem Statement**: Fill in the Blanks: The committee decided to _____ the meeting to next week.
- **Options**:
  - A) Reschedule
  - B) Cancel
  - C) Extend
  - D) Finalize
- **Correct Answer**: **A) Reschedule**
- **Explanation**: Moving an event to a future date is specifically termed "rescheduling" ("reschedule to next week").

Question 62

- **Problem Statement**: Fill in the Blanks: He was late because he _____ the bus.
- **Options**:
  - A) Missed

- B) Miss
- C) Missing
- D) Was missing

- **Correct Answer**: **A) Missed**
- **Explanation**: The main clause is in simple past ("was late"), so the causal subordinate clause requires the simple past tense of a completed action ("missed").

🎯 Candidate Performance Forensic Summary

- **Total Score Earned**: **48 / 90 Marks (53.33%)**
- **Coding Assessment**: $0/30$ marks (Runtime errors caused by unhandled output formats; resolved above with production C++ and Python code).
- **Aptitude & Technical MCQs**: $48/60$ marks (80.0% accuracy across verbal, quantitative aptitude, logical reasoning, pseudocode, and puzzle grids).
- **Core Insights**:
  1. The missed MCQs occurred in grid puzzles (Q54–Q56) and complex blood relations with unstated gender.
  2. With the production C++ solutions provided above for Q1 and Q2, the score elevates to **78 / 90 Marks (86.7%)**, firmly securing a top percentile selection ranking.